Real-time IoT networking on the Internet

Where the *physical* world meets the *virtual* world

Tom Herbert <tom@quantonium.net>



QUANTONIUM

This talk is...

- Not about XDP
- Not about BBR
- Not about netfilter or tc
- Not about busy polling, packet steering, or zero-copy
- Not even about any particular piece of code

… but it is about the applying the system to an interesting use case for which the above should be relevant

QUANTONIUM

Latency continuum (revisited)

0 msecs 1 msecs 10 msecs 100 msecs

?



Datacenter



Real-time Internet



Internet

Latency continuum (revisited)

0 msecs | 1 msecs | 10 msecs | 100 msecs



Datacenter



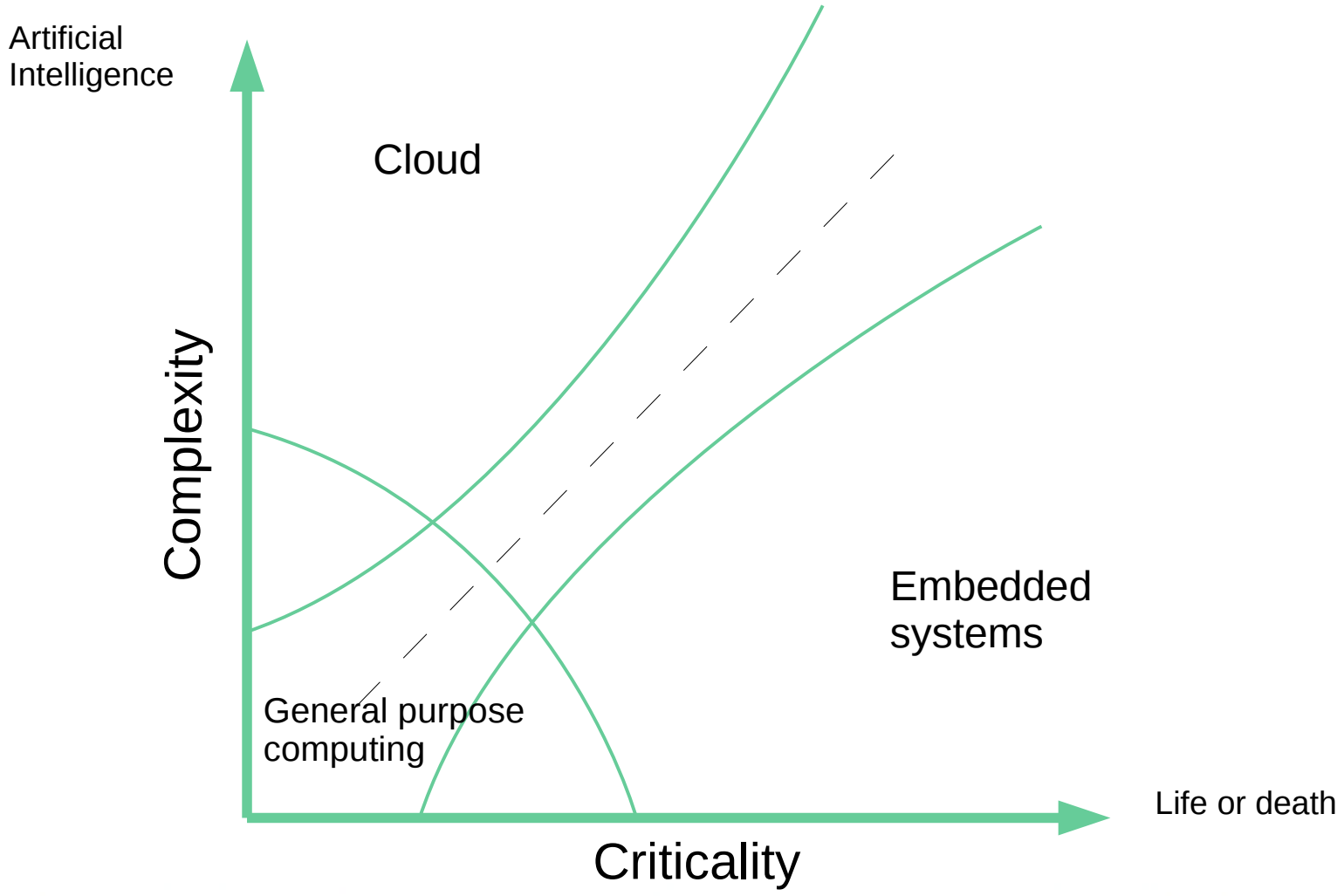
Real-time Internet



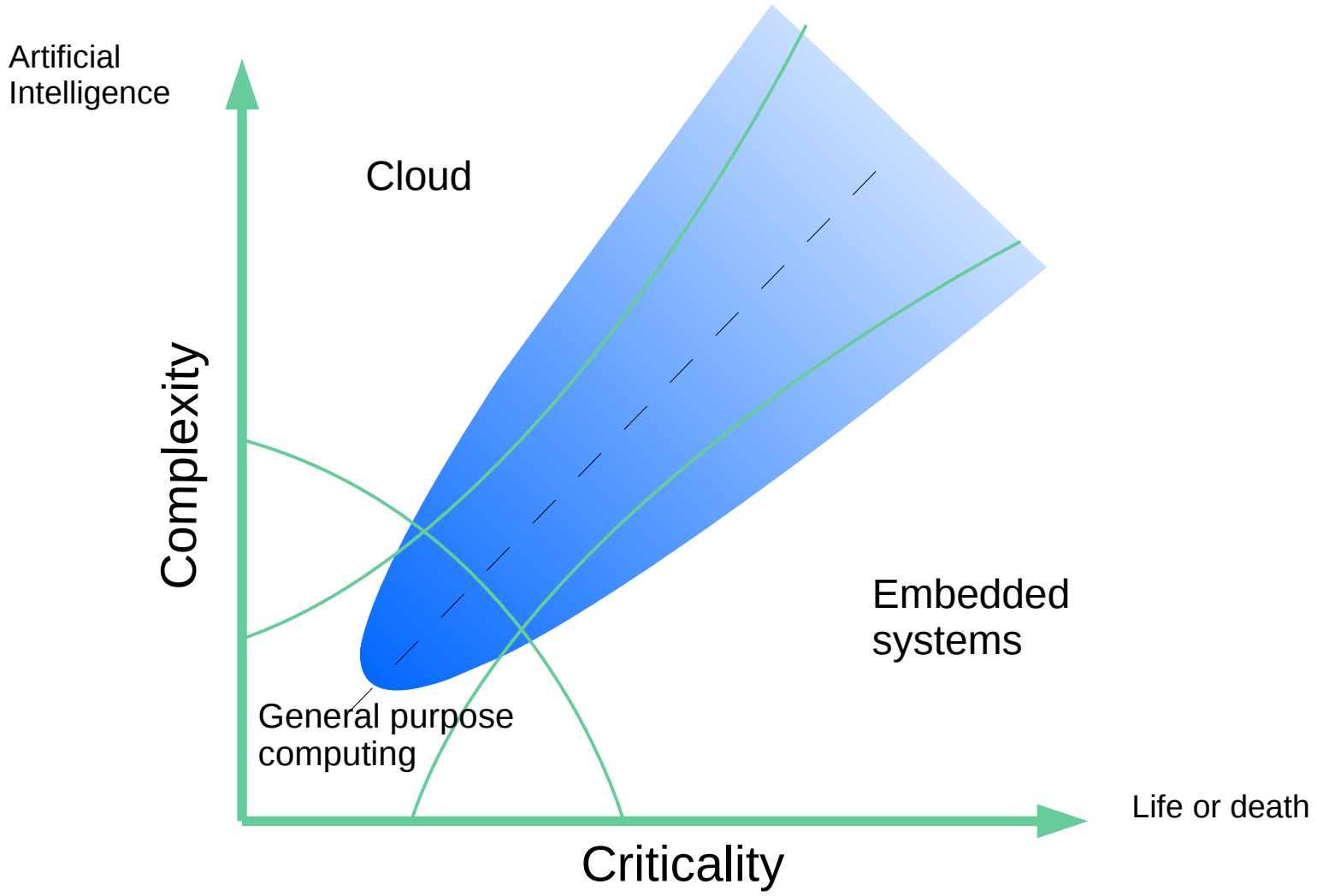
Internet

?

Motivation: the problem space



Motivation: the problem space



Characteristics

- Devices deployed in field, not datacenters
- Real-time
  - 10msecs avg., 50msecs max application latency
  - Control path must be included (e.g. connection setup, security negotiation)
- Predominantly wireless and often mobile
- Highly reliable, available, and secure
- AI and ML often part of end solution

QUANTONIUM

Conclusions #1

- Devices deployed in proximity of devices
  - Speed of light: not just a good idea, it’s the law!
  - Minimal hops (like just one) means lower latency, higher reliability, and better security
  - 10-500 meters range
- Radio
  - WAVE (Wireless Access in Vehicular Environments)
  - 802.11p (802.11 OCB)
  - 5GAA
  - White space radio, or others
  - Hopefully software defined radio goes a long way
- Networking
  - Range limits concurrent devices to few 100s
  - Data rates are low Mbits (for now)

QUANTONIUM

Conclusions #2

- Networking protocols
  - Most likely IPv6 (IPv6 over Ethernet with adaptation layer)
  - IEEE may make L3-less safety protocols
  - For mobility need to consider hand-off
- Transport protocols
  - Safety protocols are deadline based, might use FEC
  - TCP and the like may be overkill, UDP likely?
- Security
  - Ipsec, DTLS likely
  - Need to obfuscate both L2 and L3 addresses with mobility
  - DOS mitigation

QUANTONIUM

Input

History of this intersection

Emergency Services

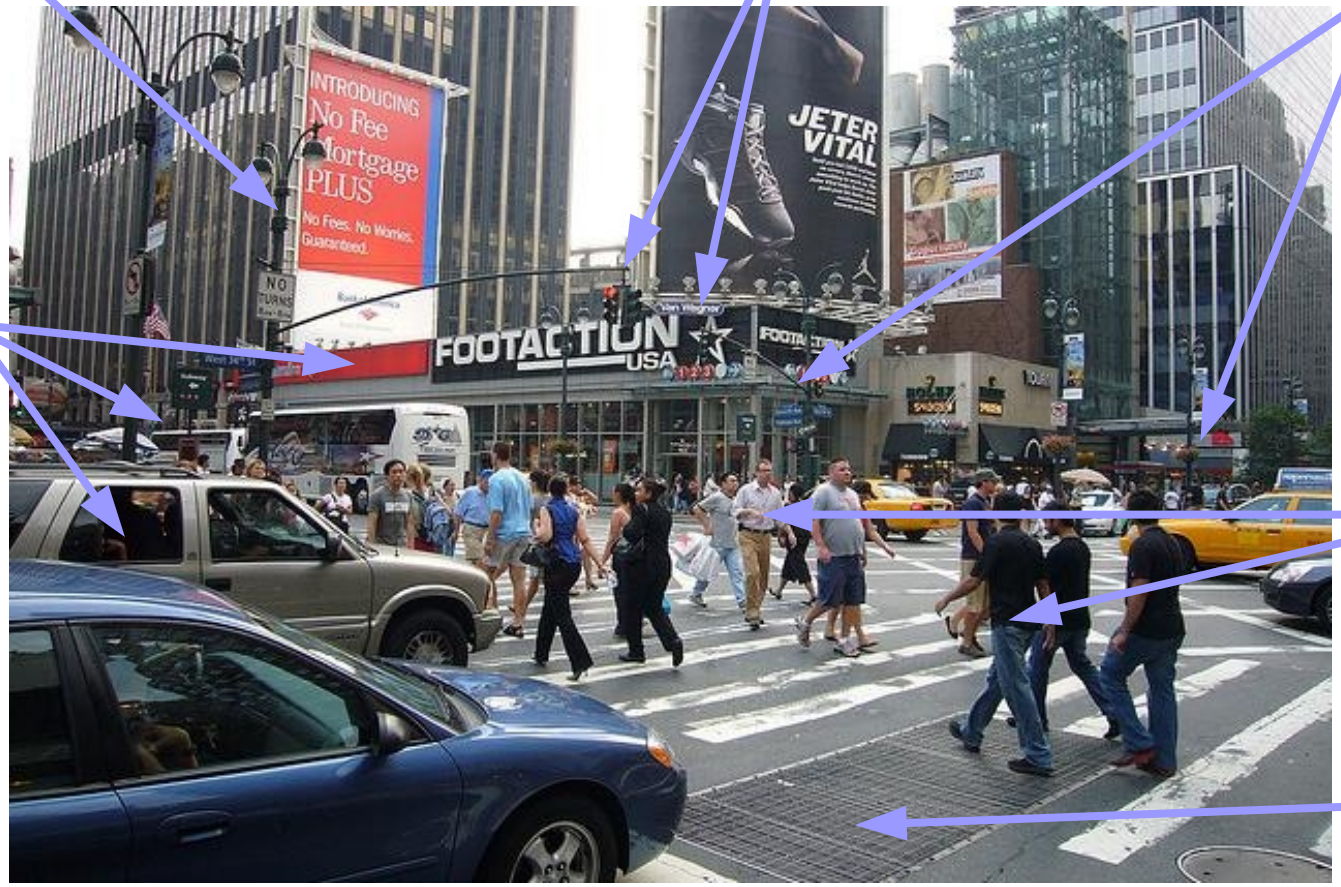
APB

Macro traffic conditions

Weather/light conditions

Traffic signals

Video, radar



Vehicles (V2X)

Personal Devices (e.g. phones)

Inductance loops

QUANTONIUM

Output

Statistics

Traffic conditions

Coordinated traffic signals

Notifications To authorities

ML feedback



Instr. to vehicles

Audio/visual warnings

Notifications to personal devices

QUANTONIUM

For Linux

- **Reliable, secure, minimal downtime upgrades**
- Device drivers for mobile
- Deadline scheduling
- Low end or high end system
  - Raspberry PI
  - Edge compute server with GPUs and gobs of memory/flash

QUANTONIUM

More on Linux

- Environment
  - Containers likely
  - OS (for IoT) beyond kernel may be substantial
- Process scheduling
  - Priority is essential
  - Real-time Linux?
- Optimizations
  - Simplicity and robustness > micro optimizations
  - User programmability (SW defined) > custom HW features
  - Upgrade-ability > raw performance

QUANTONIUM

Impact of AI

- IoT and AI (machine learning) often mentioned in same sentence
- Interesting effects on networking
  - ML to invent new CC algorithms
  - Use ML+XDP/BPF to do aggressive DDOS mitigation
  - Model communications and system after humans
- Networking interaction with the physical world
- Topic suggestion for next netdev: AI, ML, and the Linux networking stack

QUANTONIUM

Example #1: Who am I talking to?



Device gives us GPS



Of course we can’t trust device

Attacker: “Haha, I fooled you”



Triangulation and signal techniques may be harder to compromise



Correlate what we **hear** (network) with what we **see** (e.g. cameras)

Bingo!



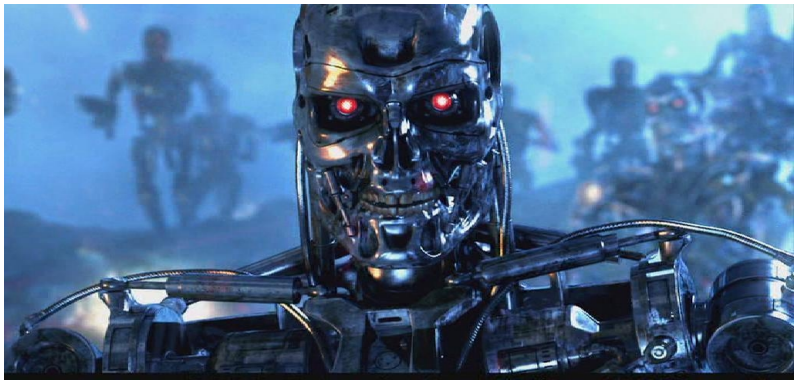
Problem #2: Teaching a sixteen year old to drive

An interesting case study in human(/machine) learning!





VS.



- A teenager in the drivers seat is a lot like current state of autonomous vehicles
- Both are “intelligent”, but in different ways (AI wins in 2029!)
- Both are Level 3 autonomy, sometimes we need to “take-over”
- Trusting our lives to them is a leap of faith
- External input mitigates above in both cases

A leisurely trip to the store:

Andy2dad protocol

00:00 Andy: *Where do I need to go?*

00:04 Dad: *Take left at light, and then first right*

00:09 Andy: *Uh-huh*

01:03 Dad: *Change lanes to turn*

01:08 Blinker on (implicit ack)

01:11 Lane change commences

-

01:34 Approaching stop sign

01:38 Dad: *There's a stop sign ahead*

01:42 Dad: *Start to stop*

01:45 Dad: *Stop!*

01:47 Dad: *STOP!*

01:47.5 Dad: **STOP!**

01:47.5 Emergency stop

A leisurely trip to the store:

Vehicle2RSU protocol

00:00.000 Vehicle: *RPC request for route from maps*
00:00.670 RSU: *RPC reply with route*
00:00.680 Vehicle: *ACK*

00:01.000 Kids playing by street
00:05.000 RSU: *Reduce speed to 30mph*
00:05.500 Vehicle: *ACK*
00:06.500 Speed reduction confirmed by RSU

00:08.500 RSU sees kid run into street
00:08.502 RSU processes input, determines decision
00:08.503 RSU: ***execute emergency stop***
00:08:510 No velocity change detected
00:08.513 RSU: ***execute emergency stop***
00:08:520 No velocity change detected
00:08.530 RSU: **execute emergency stop**
00:08:538 Velocity change detected
00:08:540- Continue sending emergency stop and tracking that car stops...
00:08:700 No collision, kid saved, no thank you though!

U

Thank you!

QUANTONIUM